# Page 6 Practical Report

Communicating Effectively with an AI Coding Agent

**Date: 21 October 2026**

## Objective

Develop a reusable prompt structure that makes an AI coding task clear, bounded, measurable, and verifiable.

## Reusable Prompt Structure

**Goal → Context → Scope → Constraints → Acceptance Criteria → Verification → Required Response**

## Weak → Better → Strong

| Level | Request |
|---|---|
| **Weak** | “Fix the filter.” |
| **Better** | “When I select Done in the status filter, Open tasks are also displayed. Expected behavior: show only Done tasks.” |
| **Strong** | “Fix the Task Tracker status filter. Scope: the filter component and relevant tests. Preserve existing All behavior; do not perform unrelated refactoring or add dependencies. Verify Open, Done, and All. Report the diff summary, assumptions, exact checks and results, and any remaining uncertainty.” |

## Copy-Ready Prompt Templates

| Task | Prompt |
|---|---|
| **Explain Code** | “Explain this feature’s data flow in simple terms. Identify relevant files and functions. List assumptions and unclear points separately. Do not modify code.” |
| **Diagnose Only** | “Compare the observed and expected behavior. Test the likely cause using evidence. Do not modify files; provide the smallest next verification step.” |
| **Bounded Change** | “Make the minimum coherent change needed to meet these acceptance criteria. Stay within the allowed scope, run relevant checks, and report the diff, results, and limitations.” |
| **Review Diff** | “Review this diff without modifying it. Check correctness, regressions, security and data access, and missing tests. Report findings with evidence and severity.” |

## Implementation Summary

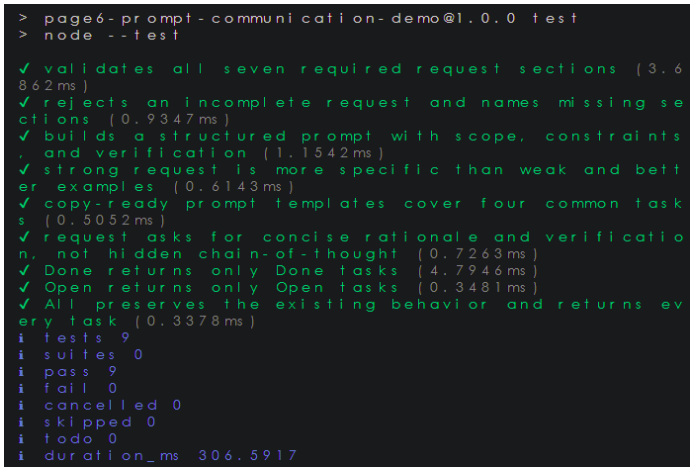
The prompt builder validates all seven required sections, identifies missing sections in incomplete requests, and generates a structured prompt. The project also includes reusable prompt templates and automated tests for request validation, prompt quality, and Task Tracker status filtering.

## Verification Results

| Total Tests | Passed | Failed | Result |
|---|---|---|---|
| 9 | 9 | 0 | **PASS** |

## Test Evidence

The terminal screenshot below records the test run: 9 tests passed and 0 failed.




## Practical Takeaway

A strong prompt defines what should change, what must remain untouched, and what evidence is needed to confirm completion.